AI-Powered Mock Interview Preparation Kit Using Generative AI and Virtual Interview Avatars

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***Abstract*—Preparation for an interview requires technical expertise, problem-solving skills, communication skills, and confidence. The conventional methods of self-preparation, static question banks, and mock interviews among peers are not personalized, scalable, or realistic. This paper proposes an AI-assisted mock interview preparation system that combines ATS-based resume evaluation, adaptive aptitude and technical quizzes, coding practice, and mock HR and technical interviews with a virtual AI interviewer.**

The proposed system utilizes the Gemini-2.5-Flash large language model for dynamic question generation, resume assessment, response evaluation, and feedback. The real-time virtual interviewer enabled by LiveKit allows for interactive mock interviews that simulate real-life situations. The experimental study with students shows enhanced readiness for interviews, boosted confidence, and progress in technical, aptitude, and communication skills, thus proving the effectiveness of the proposed system as a scalable solution for interview preparation.

***Index Terms*—Mock Interview System, Generative AI, Resume Analysis, Coding Practice, Aptitude Assessment, Virtual Interview Avatar, Gemini Model, Interview Preparation, LiveKit.**

I. Introduction

Campus placements and interviews demand technical knowledge, problem-solving skills, communication skills, and confidence. But many students lack proper interview preparation, even if they have a strong academic background. Conventional methods like static question banks, coaching classes, and friend interviews are plagued by a lack of personalization, delayed feedback, exorbitant costs, and a lack of realism.

Current breakthroughs in generative artificial intelligence and large language models make possible dynamic content creation, real-time feedback, and dialogue capabilities. Virtual avatars also add to the realism of human interviewers [16], [17].

The mock interview preparation kit that is presented in this paper is AI-enabled and has the following features:

- examines resumes to determine strengths, weaknesses, and skill gaps;
- generates adaptive aptitude and technical quizzes;
- offers coding round practice with automated evaluation;
- conducts HR and technical interviews with a virtual AI avatar.
- employs the Gemini-2.5-Flash model for intelligent content generation and feedback. The proposed system provides a unified, scalable, and realistic platform for interview preparation.

The proposed system delivers a unified, scalable, and realistic interview preparation platform tailored to individual learning needs.

II. Related Work

A. AI-Based Mock Interview Systems

There have been a few studies on the application of artificial intelligence to automate the process of mock interview generation and assessment. Li *et al.* introduced EZInterviewer, an artificial intelligence-based system for mock interview generation and automated feedback to enhance job interview skills [4]. Although useful for question-answering and assessment, the system is mainly designed for text-based conversation and lacks the multi-step preparation process of resume analysis or coding tests.

Likewise, Sharma *et al.* proposed an AI-based mock interview system that incorporates natural language processing, computer vision, and generative AI [6]. Mythili *et al.* discussed an AI-based mock interview system that focuses on automating the process of delivering interview questions and scoring

[7]. The above-mentioned systems prove the possibility of conducting interviews using AI but mostly focus on the mock interview system as a separate entity.

B. Interactive and Reflective Interview Practice

Interactive simulation and feedback-driven learning have been shown to enhance interview preparation. Daryanto *et al.* introduced Conversate, an interactive interview simulation system that supports reflective learning through dialogic feedback [5]. While effective in improving learning outcomes, such systems do not emphasize large-scale personalization or integration with technical assessments and coding rounds.

C. Virtual Avatars and AI-Generated Video

Recent advances in AI-based video generation and virtual human technologies have enabled more realistic human–computer interaction. Anitha *et al.* demonstrated the potential of dynamic AI-generated video for interactive applications [8]. These developments provide a foundation for avatar-based interview simulations; however, their application to adaptive, real-time interview preparation remains limited in existing literature.

D. Behavioral and Engagement Analysis in Learning

Behavioral analysis has been applied to assess engagement and attention in educational environments. Warankar *et al.* analyzed student attention span using deep learning techniques, highlighting the importance of behavioral cues in learning and assessment systems [9]. Although relevant, such approaches are not directly integrated into interview preparation platforms.

E. Research Gap

The existing systems are mainly centered around isolated interview preparation or assessment modules and do not have end-to-end integration. In contrast, the proposed work integrates resume evaluation, adaptive quizzes, coding exercises, and AI-powered HR and technical interviews with a virtual avatar in a single modular system. The system utilizes the Gemini-2.5-Flash intelligent content generation and evaluation model and the Livekit + beyond presence streaming avatars to fill the gaps in personalization, scalability, and comprehensive interview preparation [4]–[6].

III. System Overview

The AI-based mock interview preparation toolkit is developed as a modular, web-based platform that brings together multiple AI components into a single learning and evaluation system. It supports end-to-end interview preparation by covering resume quality, aptitude, technical knowledge, coding skills, and communication through simulated interviews.

The system consists of four tightly integrated core modules:

1) Resume Builder and Intelligent Resume Analysis
2) Aptitude and Technical Quiz Engine
3) Coding Round Practice and Evaluation Module
4) AI-Based Technical and HR Interview Module with Virtual Avatar

A centralized user profile and analytics layer connects all modules, enabling personalized content generation, adaptive difficulty adjustment, and long-term performance tracking.

A. Resume Builder and Intelligent Resume Analysis

The resume module helps users design a resume either based on structured information inputted by them or by uploading their current resume in either PDF or DOC format. The inputted resumes are processed, extracting essential information from them, particularly about education, skills, experience, etc.

Resume-job matching is evaluated through a hybrid score, similar to the ATS model. Semantic relevance is computed through a SentenceTransformer model with a bi-encoder, and keyword relevance is computed through a TF-IDF matcher. The final ATS score is a combination of the two, which mimics the actual ATS model that is commonly used in the world of resume matching. [4], [6], [11], [12]

”The system also offers suggestions on missing skills or incomplete sections and even offers explanations on ways to enhance the quality of the resume. There are also exchange mechanisms for the skill profiles between the quiz module and the interview module.”

B. Aptitude and Technical Quiz Engine

Logical reasoning skills will be evaluated and enhanced through the quiz engine. reasoning, quantitative aptitude, verbal ability and domain- special technical knowledge. Instead of static question banks, the system dynamically allows for the generation of questions using Gemini-2.5- Flash based on:

- the user’s selected job role;
- resume-derived skill profile;
- historical quiz performance.

Historical quiz performance. Each quiz session adapts in difficulty using a rule-based pro-Aggression strategy. Correct answers gradually increase question While incorrect responses trigger simpler questions, complexity and explanatory feedback [14], [15]. After each question, the systemprovides:

- correctness validation;
- step-by-step explanation;
- conceptual clarification where needed.

Quiz results are stored as time-series data, enabling longitudinal analysis of the improvement across subjects such as data structures, operating systems, database management systems,and aptitude reasoning.

C. Coding Round Practice and Evaluation Module

The coding module in PBLT will simulate real world technical coding rounds usually encountered in campus placement and industry hiring. The users are provided with programming problems of differing difficulty ranges, which may involve such areas as arrays, strings, recursion, dynamic programming, and basic algorithms.

Solutions can be provided through an in-built code editor. the system assesses the papers using a multi-stage ap- approach:

- logical correctness of the solution;

- time and space complexity analysis;
- code readability and structure.

Gemini-2.5-Flash is employed in the creation of the solutions, detect logic flaws, and give improvements suggestions. In the event that the submission was not done correctly, the system offers progressive hints rather than direct answers, encouraging learning-oriented problem solving [13].

Performance measures such as success rates, average attempts, and their topic-wise strengths, which are recorded and linked to the interview module for targeted questioning.

D. AI-Based Technical and HR Interview Module

The interview module incorporates a real-time virtual interviewer implemented with LiveKit and BeyondPresence. LiveKit provides a library for Vides low latency WebRTC-based audio-video communication tion, while the BeyondPresence avatar offers a realistic interviewer interface .

Interview questions are generated based on the Gemini-2.5-Flash model based on analysis of the resume and quiz functional performance, as well as coding results. The user responses are evaluated in real time for technical accuracy, clarity, and relevance. A The state-driven controller is responsible for the sequence of questions, ups, enabling, the timing, allowing a realistic simulation of HR and technical interviews.

The interview flow is managed by an interview state machine that manages:

- question sequencing;
- follow-up question generation;
- time management;
- interview difficulty adjustment.

Gemini-2.5-Flash generates HR and technical questions in real time based on:

- resume content;
- quiz and coding performance;
- previous interview responses.

User feedback is analyzed on technical, clarity, knowledge, confidence, and logical structure. It then assigns scores For each response, it produces a detailed feedback highlighting strengths, weaknesses, and suggested improvements.

E. Centralized Analytics and Personalization Layer

All modules feed into a centralized analytics layer, which maintains detailed performance profile for each user. This calculates metrics:

- as skill proficiency scores
- an overall interview readiness index
- & long-term improvement trends

The results are presented on interactive dashboards, giving users a clear view of their progress & helping them identify areas that need attention.

F. System Workflow Summary

The entire system works as smooth, end-to-end process:

1) User begin by uploading their resume
2) After that, it analyzes to identify key skills and areas for improvement.
3) Based on analysis, quizzes and coding challenges are generated
4) After this performance is tracked and used to adjust difficulty of interview questions
5) An AI avatar then conducts realistic HR and for technical interviews
6) Detailed feedback and insights provided from each session helps users understand their progress and guidance for next steps

Thoroughly, these components create a continuous learning loop that supports measurable improvement and makes the system well-suited for scalable use in academic and placement preparation environments.

IV. Implementation Details

The AI-based mock interview preparation toolkit is built as modular, full-stack web application. It has a hybrid architecture that brings together with Next.js frontend for the interface, FastAPI service for backend processing, and Next.js API router for seamless communication. Each major feature is developed as independent component , then integrated into a single cohesive system. This approach makes the platform easier to scale, maintain, & extend , also ensures that each part of system has clear and well-defined role.

A. Overall Architecture

The system here follows the layered architecture, with each layer for specific part of platform:

- **Presentation layer**: It is built using Next.js, to handle users interactions, dashboards, & live interview.
- **Application layer**: This layer uses Next.js API routes to manage the request handling and lightweight coordinations between services
- **AI & Processing layer**: It is powered by FastAPI & takes care of core intelligence tasks such as resume analysis, quiz generation, coding challenges, & interview evaluation
- **Data layer**: This layer manages the secure storage of user profiles, performance results, & analytical insights

Now this, architectural design allows each module to be developed and maintained independently, while still working together seamlessly as part of unified system

TABLE I
System Configuration and AI Model Parameters

Parameter	Value
Frontend framework	Next.js 14
Backend framework	FastAPI (Python 3.10)
AI model used	Gemini-2.5-Flash
Average LLM response latency	400–700 ms
Maximum prompt length	8,000 tokens
Average interview session duration	15–25 min
Concurrent users supported (test)	50+

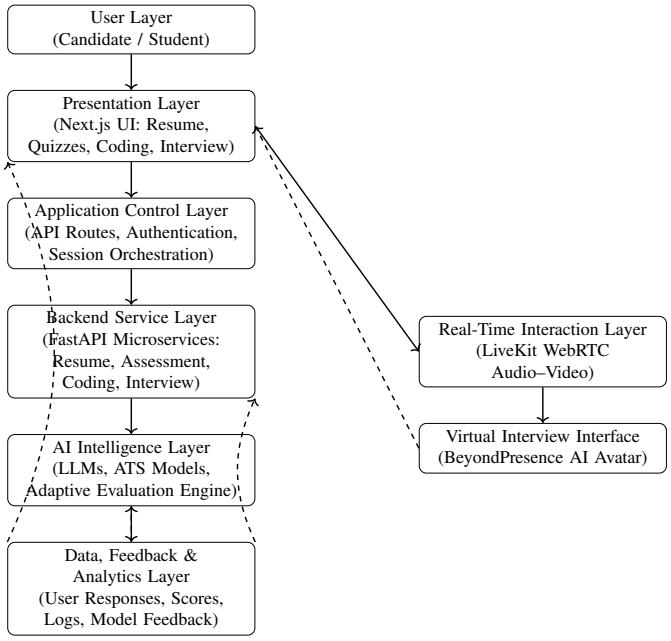

Fig. 1. Continuous Feedback-Driven Architecture of the AI-Powered Mock Interview Preparation Platform

B. Frontend Implementation Using Next.js

The Frontend has been using Next.js to take advantage of server-side rendering & component based approach, ensuring a fast, responsive, & interactive user experience. Now the key features like resume analysis, quizzes, coding challenges & interview management are organized into dedicated pages made up of reusable UI components. It utilizes React hooks and context providers for state management, allowing for easy sharing of user profiles, session data, and performance metrics among modules. This will provide a seamless interface while transitioning between the phases of preparation and an end-to-end seamless user experience.

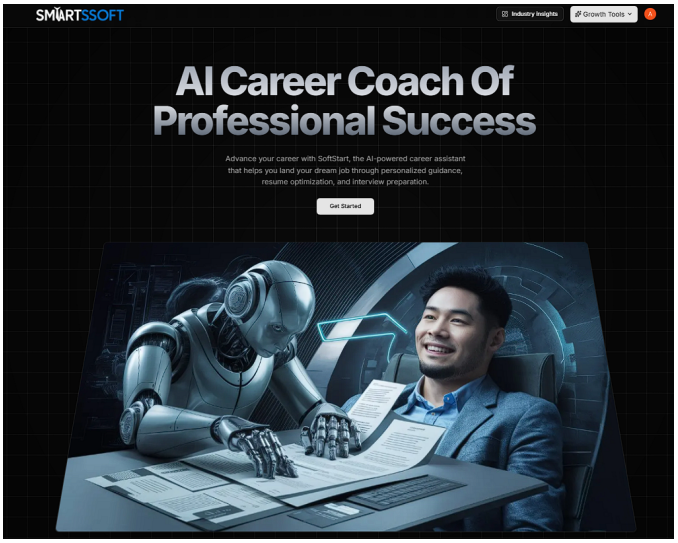

Fig. 2. Landing Page of the AI Career Coach Platform

State management is handled using React Hooks & context providers, allowing user profiles details & progress data to be shared efficiently across different modules of the application. This ensures that information stays consistent & upto date as users move through the system.

C. Backend Services Using FastAPI

Now, the FastAPI is used to power the core AI-driven backend services because of its strong support for asynchronous processing & RESTful APIs. Each major functionality is implemented as separate FastAPI service:

- like resume parsing & semantic analysis
- as aptitude & technical question generation
- the coding answer evaluation with feedback
- & the interview response analysis and scoring

Now this modular design allows FastAPI services to communicate efficiently with Gemini 2.5 Flash model using well-structured prompts & clearly formatted responses, ensuring reliable & consistent AI outputs.

TABLE II
API AND BACKEND SERVICE CHARACTERISTICS

Service Metric	Value
Average FastAPI response time	150–300 ms
Next.js API route latency	50–100 ms
Maximum request payload size	2 MB
Authentication token expiry	24 hours
Database write latency	$<$ 50 ms

D. Next.js API Routes for Module Integration

The Next.js API routes serve as bridge between frontend & FastAPI backend services. They take care of essential tasks such as, standardizing the data payloads across different modules, combining responses from multiple backend services & managing errors with appropriate fallback handling:

- that gives authentication & validation
- standardizing data payloads across different modules
- combining responses from multiple backend services
- & managing errors with appropriate fallback handling

By handling these responsibilities, the API layer hides backend complexity from the Frontend & allows backend services to be updated, replaced or extended without affecting user interface or overall user experience.

E. Resume Analysis Pipeline

When a user uploads or creates a resume, the frontend sends the resume to a Next.js API route, which in turn sends the extracted text to a FastAPI-based resume analysis service. The service analyzes the resume using an ATS-inspired assessment with a hybrid semantic and keyword-based strategy.

The semantic relevance between the resume and the target job description is calculated using a SentenceTransformer bi-encoder model, while keyword relevance is calculated using TF-IDF matching. The analysis pipeline consists of the following:

- Technical and soft skills extraction;

- Resume-job relevance analysis based on roles;
- Missing or underrepresented skills detection;
- Generation of concise and explainable improvement suggestions.

The analyzed resume data is stored and made accessible to the quiz and interview components for personalized assessment and targeted interview question generation.

F. Quiz Generation and Evaluation

The quizzes in the module are generated dynamically based on a request-response cycle. The FastAPI quiz service makes use of the Gemini-2.5-Flash library, which generates questions with predefined difficulty tags. User responses will be evaluated on the fly, along with dynamically generated explanations, reinforcing various learning aspects.

The analytics layer keeps updating and feeding performance metrics continuously, such as accuracy, response time, and topic-wise proficiency.

G. Coding Round Evaluation

The programming component utilizes an online code editor integrated with the Next.js frontend, whereby the codes entered by the user are routed to the FastAPI-based "Coding Evaluation Service" via the Next.js API routes. The codes are evaluated/analyzed :

- logical correctness;
- algorithmic efficiency;
- coding style and best practices.

Gemini-2.5-Flash generates optimized solutions and improvement feedback, which are returned to the frontend for display.

H. Virtual Interview Implementation Using LiveKit

The Live Kit is adopted for the establishment of the real-time WebRTC-based audio-video communication in this virtual interview module [2]. An avatar from Beyond Presence actually plays the role of a virtual interviewer who synchronizes with AI-generated prompts and speech output [3].

The interview prompts are generated by the Gemini-2.5-Flash model and delivered using a text-to-speech pipeline. User responses are captured and analyzed asynchronously for their content quality and relevance. This will enable low latency interaction, scalability across concurrent sessions, and realistic interview simulations [18].

I. Data Management and Analytics

The user information, session results & performance data are stored in centralized database. An analytics layer processes this data:

- to calculate skill-specific performance metrics
- the overall interview readiness scores
- & improvement trends overtime

These insights are then displayed on Frontend through interactive charts & concise summary reports, helping users clearly understand their progress & areas for improvement.

J. Security and Scalability Considerations

The API communication is secured using authentication tokens, while sensitive AI keys & configuration details are safely managed through environment variables. The modular FastAPI based architecture also supports horizontal scaling, allowing individual services to scale independently & ensuring the system remains responsive even as user demand increases.

V. Experimental Evaluation

A. Experimental Setup

The proposed system was tested with undergraduate students preparing for campus placement interviews in technical fields such as software engineering & frontend development. Students interacted with the platform across multiple sessions, including resume analysis, quizzes, coding assessments & AI-driven mock interviews. The evaluation was conducted in a controlled test environment that included the following components:

- Web-based access through modern browsers;
- Gemini-2.5-Flash model for AI processing;
- Livekit for streaming audio video interview;
- Analytics tracking.

B. Evaluation Metrics

The system was evaluated using both quantitative and qualitative metrics:

- quiz score improvement over time;
- number of questions practiced;
- coding round success rate;
- interview response quality scores;
- user confidence and satisfaction feedback.

TABLE III
Resume Analyzer Output Metrics

Metric	Observed Value
Semantic similarity score	26.16
Keyword relevance score	3.53
Final ATS score	52.25
Detected resume sections	4 / 4
Missing skills identified	1–3 per resume

TABLE IV
Observed Performance Metrics

Metric	Observed Value
Average quiz score improvement	20–25%
Coding success rate increase	25–30%
Interview response quality gain	30–35%
User-reported confidence improvement	High
Average system response time	$<$ 1 second
Interview completion rate	$>$ 90%

C. Dashboard-Based Performance Analysis

The graphical dashboard shows the progress of the user, which includes average scores, the latest performance, and the volume of practice. The performance trend graph shows the improvement of the user in the quiz sessions.

TABLE V
QUIZ AND CODING EVALUATION PARAMETERS

Metric	Configured Value
Quiz difficulty levels	Easy, Medium, Hard
Questions per quiz session	10–15
Time limit per question	60–90 seconds
Coding problem difficulty range	Easy to Medium
Maximum code submission attempts	3
Coding score weight (interview)	40%

TABLE VI
INTERVIEW PREPARATION DASHBOARD METRICS

Metric	Observed Value
Average assessment score	57.5%
Total questions practiced	40
Latest assessment score	70.0%
Assessment sessions tracked	Multiple
Performance improvement trend	Increasing

D. Virtual Interview Evaluation

The virtual interview system was a total success in simulating real-life or natural interview settings with the use of an avatar integrated with AI. The user demonstrated clear improvements in terms of the structure of responses, clarity, and self-confidence in various interview session interactions. The conversational style of the avatar eased the anxiety of the user compared to mock interviews in a neat manner

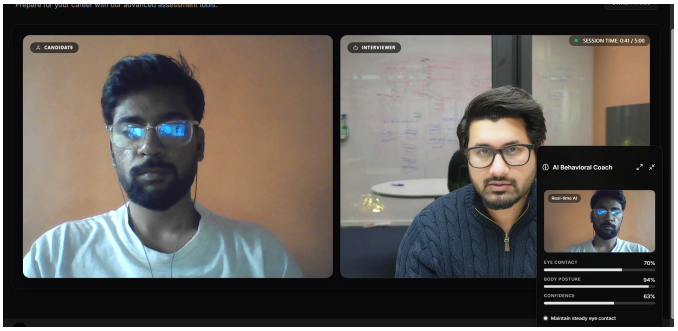

Fig. 3. Virtual Interview Conducted Using AI Avatar

TABLE VII
AI-BASED INTERVIEW SCORING WEIGHTS

Evaluation Component	Weight (%)
Technical correctness	40
Logical structure	20
Communication clarity	20
Behavioral relevance (HR)	10
Response confidence	10
Total	100

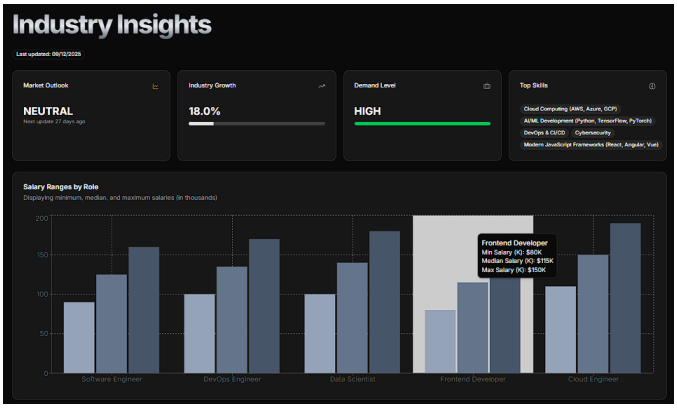

Fig. 4. Industry Insights Module with Demand and Salary Analysis

E. Industry Insights Evaluation

The industry insights module offers real-time awareness of the market with information such as the level of demand, growth rates, required skills, and salary scales. This improved awareness enabled the users to strengthen the linkage of their preparations with industry needs.

F. Observed Results

TABLE VIII
OBSERVED PERFORMANCE IMPROVEMENTS

Metric	Observed Improvement
Average quiz score	+25%
Coding problem success rate	+30%
Interview response quality	+35%
User confidence (survey-based)	High

The results indicate that the integrated AI-driven preparation workflow significantly enhances interview readiness.

VI. DISCUSSION

The experimental study indicates that the proposed AI-powered "mock interview preparation kit" effectively improves interview readiness by combining resume analysis, adaptive quizzes, coding, and AI interview situations. "The use of generative AI enables dynamic question generation" and personalized feedback, overcoming the shortcomings of static interview preparation methods. [4]–[6] Additionally, the inclusion of a virtual AI interviewer can be useful in that it can enhances realism and user engagement, while the modular Next.js, FastAPI architecture provides scalability and maintains – tainability. These findings suggest that the system is well- suited both for individual preparation and for its extension to larger recruitment workflows.

VII. LIMITATIONS AND FUTURE SCOPE

Although the proposed system demonstrates strong effectiveness in preparing for individual interviews, there are some points that need to be improved to increase its usability and effectiveness.

The current drawbacks of the proposed system are as follows:

- The proposed system uses text and structured response analysis, which is not strong in emotional or sentiment analysis.
- The proposed system does not support multilingual interviews.
- The proposed system does not allow customization of interview processes for recruiters.

Future enhancements of the proposed system will be developed on the following directions:

- **Company Oriented Virtual Placement Drives**: The system can be enhanced to provide AI-based virtual placement drives for automated candidate screening and interviewing.
- **Recruiter Customization and Analytics**: Recruiters may configure role-specific interview workflows and evaluation metrics.
- **Multimodal Interview Evaluation**: Speech and facial analysis can be incorporated to enhance behavioral assessment.
- **Multilingual and Global Interview Support**: Multilanguage interview support can improve accessibility for diverse user groups.
- **Adaptive Learning and Personalization**: Interview difficulty and feedback can be personalized using long-term performance data.

VIII. Conclusion

The content of this research was based on the design of an artificial intelligence-based mock interview preparation kit, where there was integration of resume analysis, aptitude, and technical quizzes, coding, AI-based HR, and technical interviews. The AI-based system is designed in Next.js, FastAPI, Gemini-2.5-Flash, LiveKit, and AI-based HR, technical interviews, resume analysis, coding, and aptitude quizzes.

Furthermore, it shows that there is an evident improvement in interview readiness, technical competency, and user confidence when experimentally evaluated. Moreover, the framework that this paper has proposed shows that there is an evident application of generative AI in interview preparation, which can serve as a foundation for future development of virtual recruitment/placement tools [21], [22].

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